CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Seventh Semester of B. Tech. Examination (EC)

May 2014

EC 402 RF & Microwave Engineering

Date: 20.05.2014, Tuesday **Time: 1.30 p.m. To 4.30 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

Q-1 Chose appropriate option for question (b) to (g)

(a) In a quarter-wave transformer, input impedance is equal to ________ **1**

(b) Voltage reflection coefficient is the ratio of ________ wave to _______ wave. **1**
(a)Incident, reflected (b)Reflected, incident (c)Incident, absorbed (d) Absorbed, incident

(c) The Smith chart represents a family of circles obtained by varying normalized resistance in the range of __________. **1**
(a)−1 to 0(b) 0 to 1 (c) 0 to ∞ (d) −∞ to ∞

(d) A reflex Klystron function as **1**
(a) Microwave Oscillator (b)amplifier(c)Phase shifter(d)Both amplifier and phase shifter

(e) Klystron operates on the principle of- **1**
(a) Amplitude Modulation(b)Frequency Modulation(c)Pulse Modulation(d)Velocity modulation.

(f) A space between two cavities in two cavity klystron is **1**
(a)Drift space(b)Free space(c)Running spaced)Normal space

(g) Many circles are drown in a smith chart used for transmission line calculations. The circles are shown in Fig 1 presents **1**
(a) unit circle
(b) constant reactance circle
(c) constant resistance circle
(d) constant reflection circle

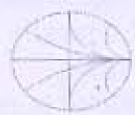

Figure 1

(h) Why TEM mode cannot propagate in Waveguide? **2**

(i) Explain standing wave in brief. **2**

Q -2 (a) Derive the wave equation with complete solutions regarding wave propagation in rectangular waveguide for TM mode **10**

(b) An air filled hollow rectangular waveguide has cross sections of $7\times3.5\ cm^2$. Calculate cutoff frequency for dominant mode? **2**

OR

Q-2 (a) Use Smith chart to find the following quantities for the transmission-line circuit shown in the Fig 2. **10**
(a) The SWR on the line.
(b) The reflection coefficient at the load.
(c) The input impedance of the line.
(d) The distance from the load to the first voltage minimum.
(e) The distance from the load to the first voltage maximum.

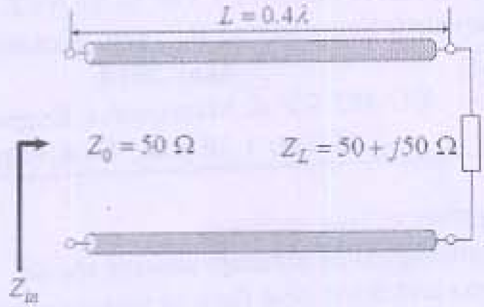

Figure 2

(b) Design a quarter wave transmission transformer to match aline having impedance of 200 ohm ,feeding at 30MHz to a load of 800ohm — 2

Q-3 (a) Based on distribution theory of transmission line, draw equivalent circuit of transmission line and equation of voltage and current for lossless transmission line. Discuss standard assumptions. — 6

(b) A rectangular air filled waveguide with dimension $7\times3.5\ cm^2$ cross is operated with a dominant mode $TE_{1,0}$ with frequency 3.5GHz. Find f_c, v_g, λ_g — 6

OR

Q-3 (a) Explain faraday rotation principle and faraday rotation isolator in detail — 6

(b) Explain operation of magic tee. — 6

Section –II

Q-4 (a) Match the following — 5

	X		Y
(a)	Hollow metal conducting pipes	(1)	Cavity resonator
(b)	interconnect two sections of waveguide	(2)	circulator
(c)	acts as a high-Q resonant circuit	(3)	PN junction
(d)	three-port microwave device used for coupling energy in only one direction	(4)	T section
(e)	diodes is not typically used in the microwave region	(5)	TWTA

(b) Why strapping is required in magnetron? — 2

(c) What structure in the TWT delays the forward progress of the traveling wave? — 1

(d) Explain electron bunching process with help of diagram — 3

Q-5 (a) Explain two cavity klystron amplifier in detail. — 6

(b) A microstripline has following parameters ε_r=5.23,h=0.1778mm,t=0.07112mm,w=0.254mm,calculate z_0 — 2

(c) A transmission line has characteristic impotence of 50 ohm and is terminated in a load impedance of 73 ohm. Calculate reflection coefficient and standing wave ration. — 4

OR

Q- 5 (a) Explain microstrip line and strapline in brief — 6

(b) Draw appropriate diagram of TWTA and explain amplification process along with diagram. — 6

Q-6 (a) Explain waveguide bend, twist and corner with the help of diagram — 6

(b) Explain directional coupler in detail — 6

OR

Q- 6 (a) Explain Parametric up-converter and parametric down-converter with the help of equations. — 6

(b) What is tunneling? Explain energy band diagram and IV characteristics of tunnel diode. — 6